

Dates -
Scales
Work plan and names

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<https://www.kaggle.com/datasets/janus137/mlb-postseason-2025-pitch-by-pitch-data>

MLB postseason pitch by pitch data

Examine the pitch by pitch data from the 2025 MLB postseason to analyze several questions/make predictions about postseason pitching

1. "What underlying factors explain the variance in pitch characteristics, and how do pitch characteristics relate to pitch/at bat outcomes?" **(RONAN)**
 - a. Using PCA + Clustering to identify what kind of pitchers are the same, what kind of pitchers are the same and how we can group them
 - b. Also be able to possibly identify their pitcher profiles with their playoff success
 - i. I.e. look at the pitching profiles and ERA and compare/ build a regression model to see if there is a relationship between pitching profile and ERA/ other pitching statistic
 - c. Variables that define a pitch, are things such as vertical and horizontal break, velocity, spin rate, extension, etc
 - d. Outcomes could be measured in several ways, possibly delta win probability or xWOBA(expected weighted on base average) of batted ball
 - e. Purpose: Analyzing the variance in pitch characteristics will allow us to derive insights to optimal pitching strategies and player evaluation
2. Can we predict if a batter will swing and miss at a pitch? **(DANIEL)**
 - a. I.e. Given a pitcher's pitch to the plate, find out if the batter will swing and miss.
 - b. Possibly only examine pitches where the batter did swing, or rather just predict for a given pitch will have the hitter both swing and miss
 - c. Neural network or logistic regression model to see how accurately we can predict if a hitter will swing and miss at a given pitch.
 - d. Variables to be considered should be count, previous pitches, spin rate, velocity, extension, previous pitches location, handedness...etc.
 - e. Possible prior clustering of pitchers and hitters into different groups based on pitching/hitting style, or inputting season level stats as features to further enhance prediction
 - f. Analyze model with ROC-AUC to test how accurate it is
 - g. Purpose: Analyzing the model will help us understand what pitch attributes and sequencings are most conducive to swing and misses

Sun	Mon	Tue	Wed	Thu	Fri	Sat
			1	2	3	4
				TODAY		
5 Retrieve Complete Dataset	6	7 Generate Cluster Groups for pitches (RONAN)	8 Figure out features for model (DANIEL)	9 Generate Cluster Groups for Pitchers (RONAN)	10	11 Run Model/ trial and error (DANIEL)
12	13 Tune model and hyperparam eters (DANIEL)	14 Model final touches and get findings (RONAN + DANIEL)	15 Finish Question 2 Finish Question 1	16	17	18
19 Modelling Done	20 Build Presentation Template	21	22	23 Upload Models	24 Interpret Models	25 Final Touches to Presentat ion
26 Presentation Completed	27 Presenting Day	28	29	30		